

ME 350 Lecture #7

DC Motors

- DC Motor Model
- Motor power and efficiency
- Motor limits (constraints)
- Optimal Motor Design

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DC Motor: Basic Equations

If a current is applied to the armature, a torque will be generated as a function of the motor construction,

Motor Torque : $T = K_T i_m$

where

T	motor torque
K_T	torque constant based on the motor construction
i_m	armature current

Analogous, if the motor turns a voltage will be created. This device is commonly termed a generator.

Electromotive Force: $EMF = V_b = K_b \omega$

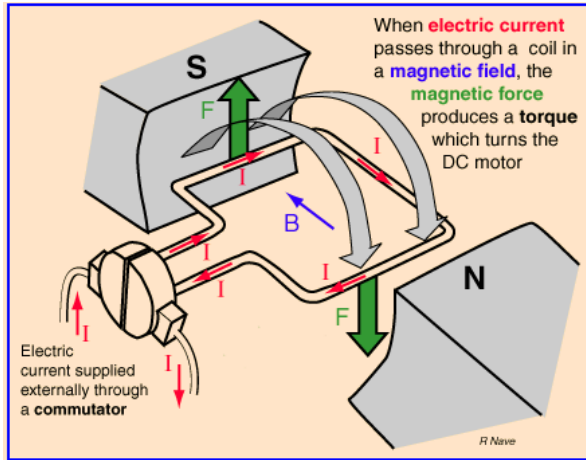
where

V_b	voltage generated by the turning motor
K_b	back emf constant based on the motor construction
ω	Speed of the motor (rpm)

Torque and back emf constants are equal and depend on the motor size and design parameters (number of coils, size of rotor windings, strenght of magnetic, etc):

$$K_T = K_b = K$$

DC Motor Model Worksheet



- T_m : Motor Output Torque
- ω : Motor Output Speed
- V_s : Supply voltage
- i_m : Motor Winding Current
- R : Winding Resistance
- L : Winding Inductance
- V_b : Back EMF
- V_R : Voltage across resistor
- V_L : Voltage across inductor
- K_t : Motor's torque constant
- K_b : Motor's back EMF constant
- T_s : Stall torque
- K : Torque-speed gradient

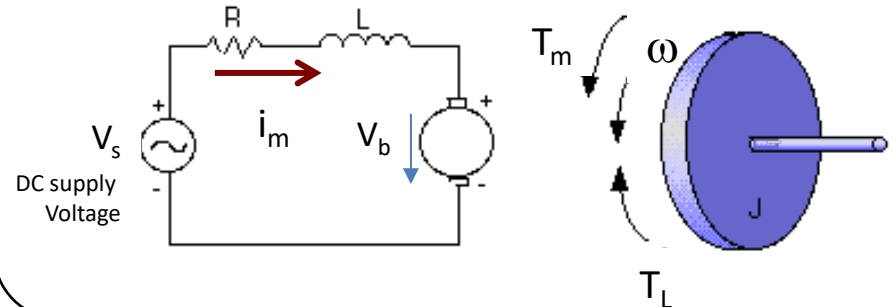
Note that $K_t = K_b$ (if expressed in N-m/A and V/(rad/s), respectively)

Permanent Magnet Brushed DC Motor Model

Model Nomenclature

T_m :	Torque generated by motor
T_L :	Load Torque
T_s :	Stall Torque
ω :	Output speed of motor
i_m :	Motor current
V_s :	Supply voltage
V_b :	Back emf
R :	Winding Resistance
L :	Winding Inductance
K_m :	Motor constant
K_t :	Torque constant
K_b :	Back-EMF constant

Basic Motor Circuit



Electrical:

Kirchoff's Voltage Law (KVL)

$$+V_s - i_m R - L \frac{di_m}{dt} - V_b = 0$$

rewriting and grouping gives:

$$(V_s - V_b) = i_m R + L \frac{di_m}{dt}$$

Substituting in the "back Emf" relationship:

$$V_s - K_b \omega = i_m R + L \frac{di_m}{dt}$$

Mechanical:

$$T_m - T_L = J \frac{d\omega}{dt}$$

Motor Operation Regions

$$(V_s - V_b) = i_m R + L \frac{di_m}{dt}$$

$$T_m - T_L = J \frac{d\omega}{dt}$$

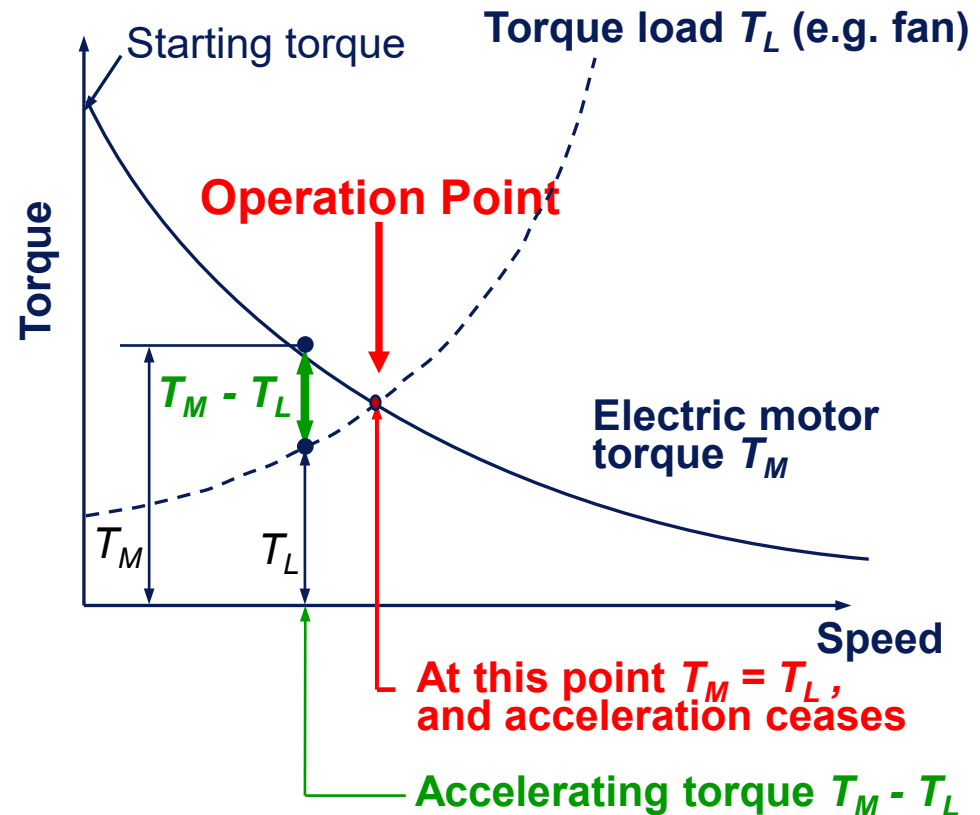
Two main operations regions:

I. Start Up Torque

This is the torque required to change a speed in a motor. Normally used for starting a motor from a dead stop. Inertia and response time is important.

II. Operating Torque

This is the torque required to operate the motor at a constant speed. Inertial loads are zero at constant speeds.



What Happens if the Motor torque (T_m) is **greater** than the Load torque (T_L)?

What Happens if the Motor torque (T_m) is **less** than the Load torque (T_L)?

Motor Operation Regions

$$(V_s - V_b) = i_m R + L \frac{di_m}{dt}$$

$$T_m - T_L = J \frac{d\omega}{dt}$$

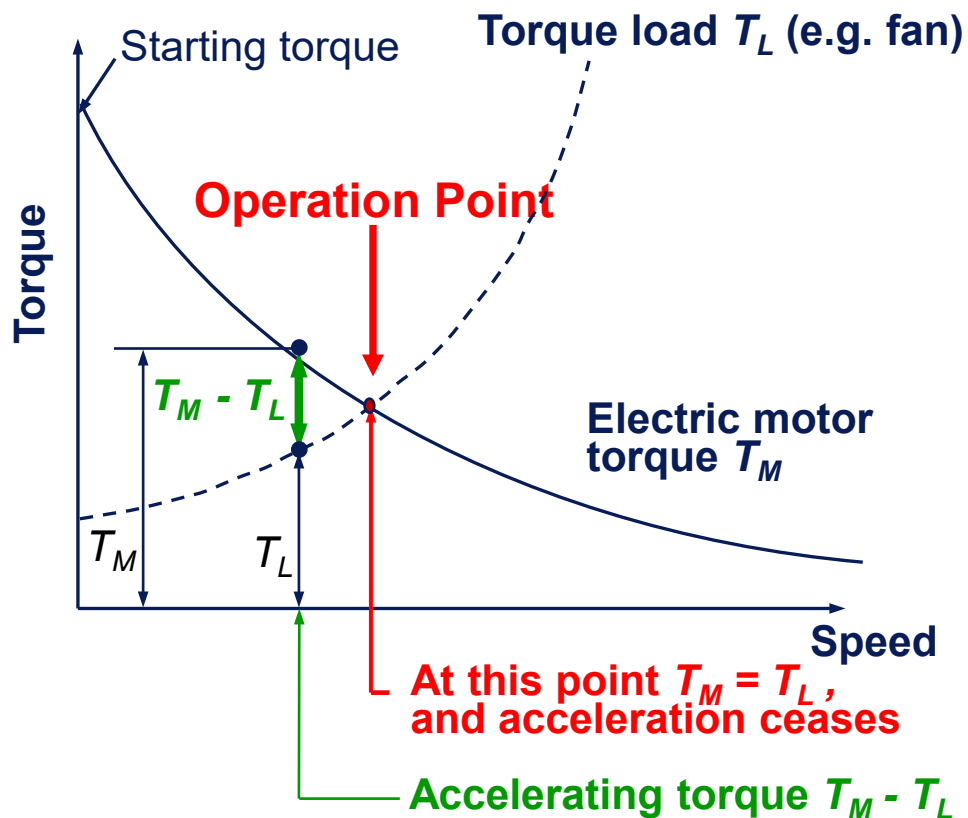
Two main operations regions:

I. Start Up Torque

This is the torque required to change a speed in a motor. Normally used for starting a motor from a dead stop. Inertia and response time is important.

II. Operating Torque

This is the torque required to operate the motor at a constant speed. Inertial loads are zero at constant speeds.



If $T_m (= K_T i_m) > T_L$

- Motor shaft will accelerate, ω goes up
- Back emf will increase causing voltage difference $(V_s - V_b)$ to decrease
- Causes current to decrease
- T_m decreases (until it reaches T_L)

If $T_m (= K_T i_m) < T_L$

- Motor shaft will decelerate, ω goes down
- Back emf will decrease causing voltage difference $(V_s - V_b)$ to increase
- Causes current to increase
- T_m increases (until it reaches T_L)

This keeps the motor stable, driving to $T_m = T_L$

Electrical DC motor model: Steady State

Given: $(V_s - V_b) = i_m R + L \frac{di_m}{dt}$

At steady-state:

- $T_m = T_L$ (net torque on motor is zero)
- $\omega = \text{constant}$
- $\frac{di_m}{dt}$ derivative goes to zero

$$V_s - V_b = i_m R$$

Substituting in $T_m = K_T i_m$ and $V_b = K_b \omega$

$$V_s - K_b \omega = i_m R = \frac{T_m R}{K_T}$$

Rewriting

$$T_m = \frac{V_s K_T}{R} - \frac{K_T K_b}{R} \omega$$

this can be simplified to

DC steady-state model

$$T_m = T_s - K_m^2 \omega$$

where *stall torque* is defined as:

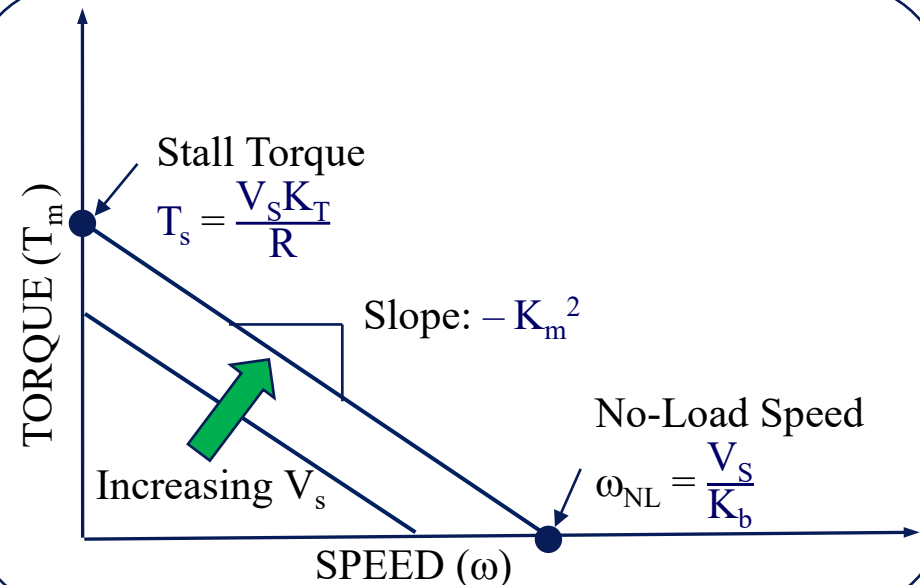
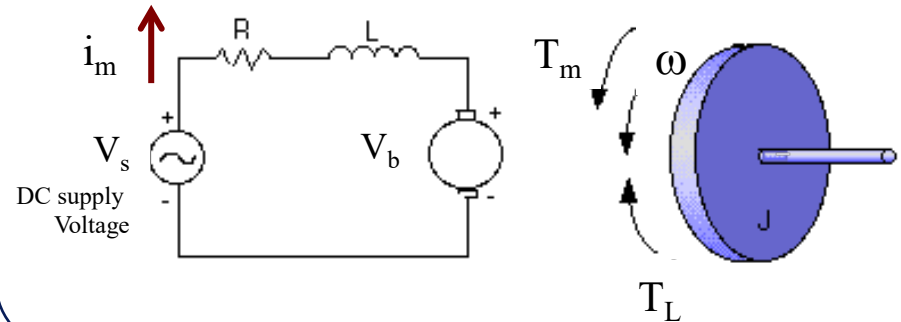
$$T_s = \frac{V_s K_T}{R}$$

If $T_L > T_s$ the motor will not move

and *motor constant* is defined as:

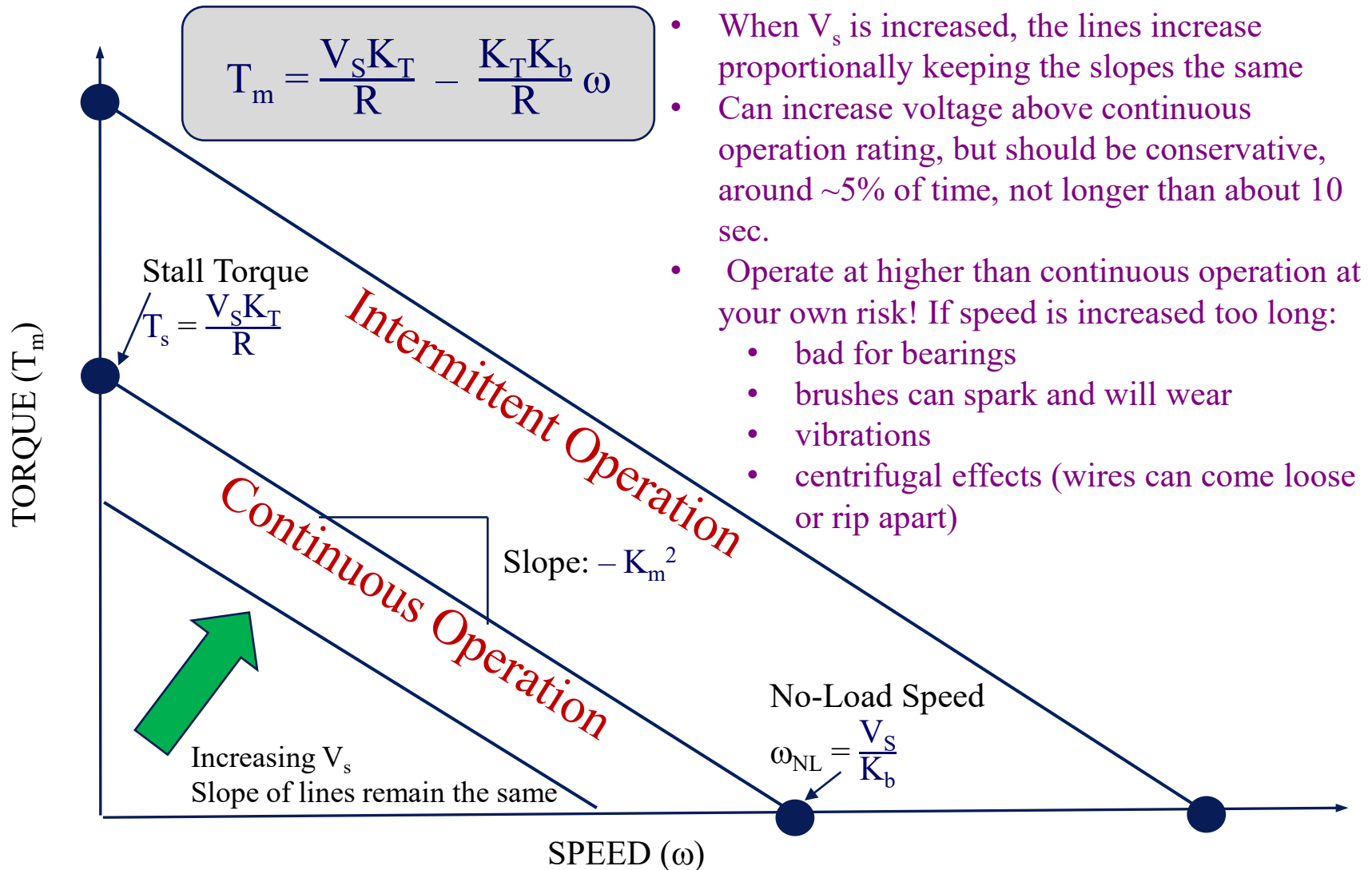
$$K_m = \sqrt{\frac{K_T K_b}{R}}$$

Basic Motor Circuit



Note: K_m is motor constant. Sometimes equation is given with “gradient” K as the slope which is not squared. Always check the units

DC Electrical Motor Curves



- When V_s is increased, the lines increase proportionally keeping the slopes the same
- Can increase voltage above continuous operation rating, but should be conservative, around ~5% of time, not longer than about 10 sec.
- Operate at higher than continuous operation at your own risk! If speed is increased too long:
 - bad for bearings
 - brushes can spark and will wear
 - vibrations
 - centrifugal effects (wires can come loose or rip apart)

How to read Motor Specs



Assembly Data	Symbol	Units	Value	
Reference Voltage	E	V	24	
No-Load Speed	S_{NL}	rpm (rad/s)	5,331	(558)
Continuous Torque (Max.) ¹	T_C	oz-in (N-m)	11.5	(8.1E-02)
Peak Torque (Stall) ²	T_{PK}	oz-in (N-m)	77	(5.4E-01)
Weight	W_M	oz (g)	19	(524)
Motor Data				
Torque Constant	K_T	oz-in/A (N-m/A)	6.00	(4.24E-02)
Back-EMF Constant	K_E	V/krpm (V/rad/s)	4.44	(4.24E-02)
Resistance	R_T	Ω	1.85	
Inductance	L	mH	1.97	
No-Load Current	I_{NL}	A	0.18	
Peak Current (Stall) ²	I_P	A	13.0	
Motor Constant	K_M	oz-in/ $\sqrt{W}$ (N-m/ $\sqrt{W}$)	4.41	(3.11E-02)
Friction Torque	T_F	oz-in (N-m)	0.80	(5.6E-03)
Rotor Inertia	J_M	oz-in-s ² (kg-m ²)	1.2E-03	(8.5E-06)
Electrical Time Constant	τ_E	ms	1.06	
Mechanical Time Constant	τ_M	ms	8.9	
Viscous Damping	D	oz-in/krpm (N-m-s)	0.055	(3.7E-06)
Damping Constant	K_D	oz-in/krpm (N-m-s)	14	(9.7E-04)
Maximum Winding Temperature	θ_{MAX}	°F (°C)	311	(155)
Thermal Impedance	R_{TH}	°F/watt (°C/watt)	52.2	(11.2)
Thermal Time Constant	τ_{TH}	min	13.8	

Exercise 1

For the highlighted motor, calculate T_s (stall torque) and i_s (starting/stall current) from the provided parameters.

Assume that the motor operates at nominal voltage

			118693	118694	118695	118696	118697	118698	118699	118700	118701
Motor Data											
Values at nominal voltage											
1	Nominal voltage	V	1.8	2.4	3	3.2	4.5	4.8	7.2	9	12
2	No load speed	rpm	4990	6360	6890	6270	6740	5700	6890	6740	7130
3	No load current	mA	23.5	25.4	23	18.6	14.8	10.8	9.57	7.4	6.05
4	Nominal speed	rpm	4320	5510	5820	4930	5050	3630	4810	4630	5030
5	Nominal torque (max. continuous torque)	mNm	2.39	2.5	2.89	3.41	4.48	5.61	5.54	5.48	5.48
6	Nominal current (max. continuous current)	A	0.72	0.72	0.72	0.72	0.72	0.711	0.566	0.438	0.348
7	Stall torque	mNm	15.5	16.9	17.3	15.2	17.4	15.2	18.1	17.4	18.6
8	Starting current	A	4.53	4.71	4.19	3.13	2.74	1.9	1.82	1.37	1.16
9	Max. efficiency	%	86	86	86	85	86	86	86	86	86
Characteristics											
10	Terminal resistance	Ω	0.397	0.51	0.715	1.02	1.64	2.53	3.95	6.56	10.3
11	Terminal inductance	mH	0.0207	0.0227	0.0302	0.0415	0.0711	0.113	0.174	0.284	0.452
12	Torque constant	mNm/A	3.43	3.58	4.13	4.84	6.34	7.99	9.92	12.7	16
13	Speed constant	rpm/V	2790	2660	2310	1970	1510	1190	962	753	597
14	Speed / torque gradient	rpm/mNm	323	379	400	415	391	378	383	389	386
15	Mechanical time constant	ms	5.84	5.71	5.56	5.46	5.36	5.31	5.29	5.29	5.27
16	Rotor inertia	gcm ²	1.73	1.44	1.33	1.26	1.31	1.34	1.32	1.3	1.3



Exercise 1: Solution

For the highlighted motor, calculate T_s (stall torque) and i_s (starting/stall current) from the provided parameters.

Assume that the motor operates at nominal supply voltage

$$T_s = \frac{K_t V_s}{R} = \frac{16 \text{ mNm} \times 12 \text{ V}}{10.3 \Omega = \text{V/A}} = 18.64 \text{ mNm}$$

$$T_s = 18.64 \text{ mNm}$$

$$T_m = K_t i_m \Rightarrow i_m = \frac{T_m}{K_t} \Rightarrow i_s = \frac{T_s}{K_t} = \frac{18.64 \text{ mNm}}{16 \text{ mNm/A}} = 1.165 \text{ A}$$

Similar to numbers in table

Motor Data				118701
Values at nominal voltage				
1	Nominal voltage	V	12	
2	No load speed	rpm	7130	
3	No load current	mA	6.05	
4	Nominal speed	rpm	5030	
5	Nominal torque (max. continuous torque)	mNm	5.48	
6	Nominal current (max. continuous current)	A	0.348	
7	Stall torque	mNm	18.6	
8	Starting current	A	1.16	
9	Max. efficiency	%	86	
Characteristics				
10	Terminal resistance	Ω	10.3	
11	Terminal inductance	mH	0.452	
12	Torque constant	mNm/A	16	
13	Speed constant	rpm/V	597	
14	Speed / torque gradient	rpm/mNm	386	
15	Mechanical time constant	ms	5.27	
16	Rotor inertia	gcm ²	1.3	



Exercise 2

Calculate the operating speed if the motor highlighted in the spec sheet below is used to drive a load with torque characteristics given by

$$T_L = A + B\omega, \text{ where } A = 2 \text{ mN-m and } B = 0.001 \text{ mN-m/rpm}$$

Assume that the motor operates at nominal voltage

			118693	118694	118695	118696	118697	118698	118699	118700	118701
Motor Data											
Values at nominal voltage											
1	Nominal voltage	V	1.8	2.4	3	3.2	4.5	4.8	7.2	9	12
2	No load speed	rpm	4990	6360	6890	6270	6740	5700	6890	6740	7130
3	No load current	mA	23.5	25.4	23	18.6	14.8	10.8	9.57	7.4	6.05
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7	Stall torque	mNm	15.5	16.9	17.3	15.2	17.4	15.2	18.1	17.4	18.6
8	Starting current	A	4.53	4.71	4.19	3.13	2.74	1.9	1.82	1.37	1.16
9	Max. efficiency	%	86	86	86	85	86	86	86	86	86
Characteristics											
10	Terminal resistance	Ω	0.397	0.51	0.715	1.02	1.64	2.53	3.95	6.56	10.3
11	Terminal inductance	mH	0.0207	0.0227	0.0302	0.0415	0.0711	0.113	0.174	0.284	0.452
12	Torque constant	mNm/A	3.43	3.58	4.13	4.84	6.34	7.99	9.92	12.7	16
13	Speed constant	rpm/V	2790	2660	2310	1970	1510	1190	962	753	597
14	Speed / torque gradient	rpm/mNm	323	379	400	415	391	378	383	389	386
15	Mechanical time constant	ms	5.84	5.71	5.56	5.46	5.36	5.31	5.29	5.29	5.27
16	Rotor inertia	gcm ²	1.73	1.44	1.33	1.26	1.31	1.34	1.32	1.3	1.3



Exercise 2: Solution

Calculate the operating speed if the motor highlighted in the spec sheet below is used to drive a load with torque characteristics given by

$T_L = A + B\omega$, where $A = 2 \text{ mN-m}$ and $B = 0.001 \text{ mN-m/rpm}$

Assume that the motor operates at nominal voltage

$$T_L = A + B\omega; \quad T_m = T_s - K\omega$$

At equilibrium: $T_L = T_m$, $\omega = \omega^*$

$$A + B\omega^* = T_s - K\omega^* \Rightarrow$$

$$\boxed{\omega^* = \frac{T_s - A}{B + K}} \quad \checkmark$$

From Catalog:

$$\left. \begin{array}{l} T_s = 18.6 \text{ mNm} \\ K = \frac{1}{386} \frac{\text{mNm}}{\text{rpm}} \end{array} \right\}$$

$$\Rightarrow \boxed{\omega^* = 4623 \text{ rpm}}$$

Motor Data

Values at nominal voltage		
1 Nominal voltage	V	12
2 No load speed	rpm	7130
3 No load current	mA	6.05
4 Nominal speed	rpm	5030
5 Nominal torque (max. continuous torque)	mNm	5.48
6 Nominal current (max. continuous current)	A	0.348
7 Stall torque	mNm	18.6
8 Starting current	A	1.16
9 Max. efficiency	%	86
Characteristics		
10 Terminal resistance	Ω	10.3
11 Terminal inductance	mH	0.452
12 Torque constant	mNm/A	16
13 Speed constant	rpm/V	387
* 14 Speed / torque gradient	rpm/mNm	386
15 Mechanical time constant	ms	5.27
16 Rotor inertia	gcm ²	1.3

Exercise 3

What supply voltage is needed to operate the highlighted motor at 3000 rpm if it drives the same load as in Exercise 2 (with torque characteristics given by

$$T_L = A + B\omega, \text{ where } A = 2 \text{ mN-m and } B = 0.001 \text{ mN-m/rpm})$$

		118693	118694	118695	118696	118697	118698	118699	118700	118701
Motor Data										
Values at nominal voltage										
1 Nominal voltage	V	1.8	2.4	3	3.2	4.5	4.8	7.2	9	12
2 No load speed	rpm	4990	6360	6890	6270	6740	5700	6890	6740	7130
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7 Stall torque	mNm	15.5	16.9	17.3	15.2	17.4	15.2	18.1	17.4	18.6
8 Starting current	A	4.53	4.71	4.19	3.13	2.74	1.9	1.82	1.37	1.16
9 Max. efficiency	%	86	86	86	85	86	86	86	86	86
Characteristics										
10 Terminal resistance	Ω	0.397	0.51	0.715	1.02	1.64	2.53	3.95	6.56	10.3
11 Terminal inductance	mH	0.0207	0.0227	0.0302	0.0415	0.0711	0.113	0.174	0.284	0.452
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16 Rotor inertia	gcm ²	1.73	1.44	1.33	1.26	1.31	1.34	1.32	1.3	1.3



Exercise 3: Solution

What supply voltage is needed to operate the highlighted motor at 3000 rpm if it drives the same load as in Exercise 2 (with torque characteristics given by

$T_L = A + B\omega$, where $A = 2 \text{ mN-m}$ and $B = 0.001 \text{ mN-m/rpm}$)

$$\omega^* = 3000 \text{ rpm}; T_L = A + B\omega^* = 2 + 0.001 \times 3000$$

$$\text{We want } T_m = T_L @ \omega^* \Rightarrow T_L = 5 \text{ mNm}$$

$$\Rightarrow T_s - K\omega^* = 5 \text{ mNm}$$

$$T_s = 5 \text{ mNm} + K\omega^* = 5 + \frac{1}{386} \times 3000 = 12.772 \text{ mNm}$$

$$\text{But } T_s \propto V_s$$

$$\Rightarrow \frac{V_s}{T_s} = \frac{V_{s, \text{nom}}}{T_{s, \text{nom}}}$$

$$\Rightarrow V_s = \frac{V_{s, \text{nom}} \cdot T_s}{T_{s, \text{nom}}} = \frac{12 \times 12.772}{18.6}$$

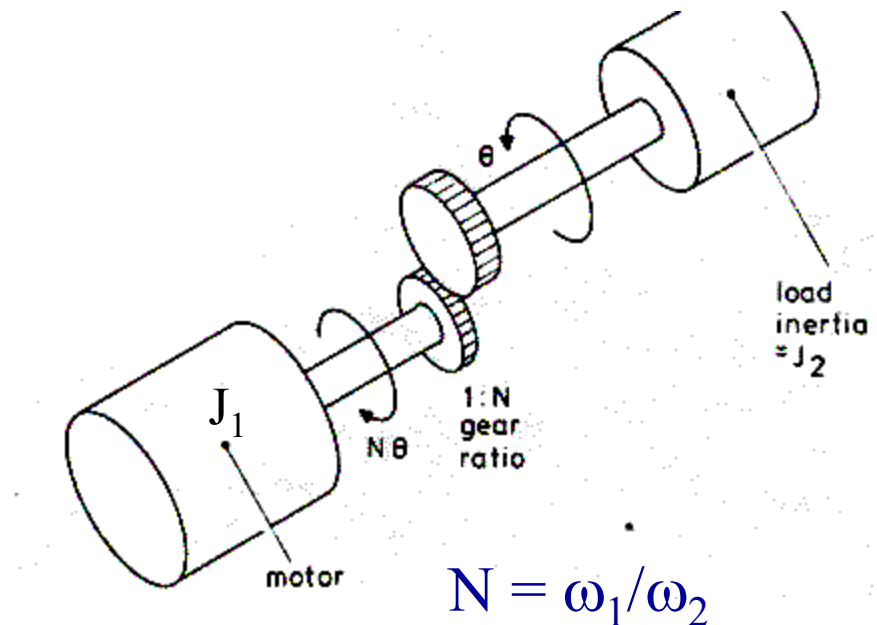
$$\Rightarrow \boxed{V_s = 8.24 \text{ V}}$$

Motor Data			118701
Values at nominal voltage			
1	Nominal voltage	V	12
2	No load speed	rpm	7130
3	No load current	mA	6.05
4	Nominal speed	rpm	5030
5	Nominal torque (max. continuous torque)	mNm	5.48
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7	Stall torque	mNm	18.6
8	Starting current	A	1.10
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15	Mechanical time constant	ms	5.27
16	Rotor inertia	gcm ²	1.3

Geared Systems, Optimum Gear Ratios, and Motor Selection

- Motors may be direct drive (motor coupled directly to load) or geared systems.
- For geared systems, choice of the motor also involves a choice of gear ratio.

Neglect backlash and elasticity in either gear teeth or shafts

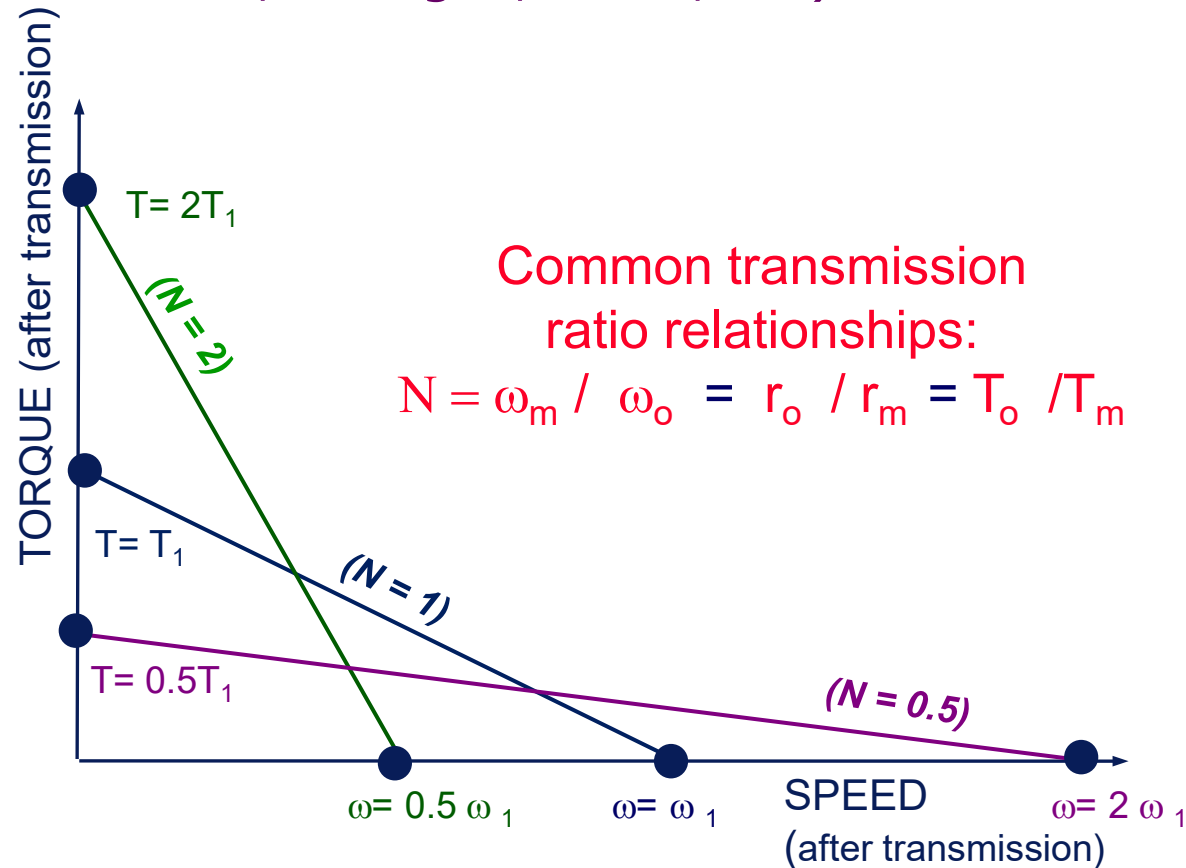


Effect of Transmission

A transmission is used to convert power from one form into another (– E.g., rotary to linear motion, low-torque high-speed to high-torque low-speed) and comes in several forms (gears, belt, linear screw drives, linkages, cams, etc).

While a transmission can manipulate the torque/speed curve to meet more the exact performance requirements of your application, the energy ideally stays the same.

*****Note: most transmissions do not have 100% efficiency, so there will be losses and thus a "cost" to using a transmission.**



Start-up Torque

When a motor changes speed, such as at start-up, it undergoes acceleration and the moment of inertia can have a significant impact. The system response time is slowed by the moment of inertia whereas the steady-state speed is unaffected. Sometimes the torque required for start-up is higher than the operating torque and is the main design driver.

Start-up Torque of Motor: $T = J \alpha_m$ or
 $T = (J_m + (J_L / N^2)) \times (N \alpha_L)$

The Moment of Inertia: $J = J_m + (J_L / N^2)$

where

J	=	Total Moment of Inertia
J_m	=	Moment of Inertia of the motor
J_L	=	Moment of Inertia of the load
J_L / N^2	=	Reflected load inertia
N	=	transmission reduction of N:1

The angular acceleration: $\alpha_m = N \alpha_L$

where α_m = angular acceleration of the motor
 α_L = angular acceleration of the load

Start-up Torque

An important consequence of this result is that motor inertia is often the dominant inertia in a servo system. Consider the following numerical example:

Motor rotor inertia $J_1 = 1$, load inertia $J_2 = 100$, and gear ratio $N = 50$.

Physically the load appears 100 times “larger” than the motor, but because of the high (but not unusual) gear ratio, the motor’s inertia effect is $N^2 J_1$ or 2,500, i.e., 25 times larger than the load.

Start-up Torque of Motor: $T = J \alpha_m$ or

$$T = (J_m + (J_L / N^2)) \times (N \alpha_L)$$

The Moment of Inertia: $J = J_m + (J_L / N^2)$

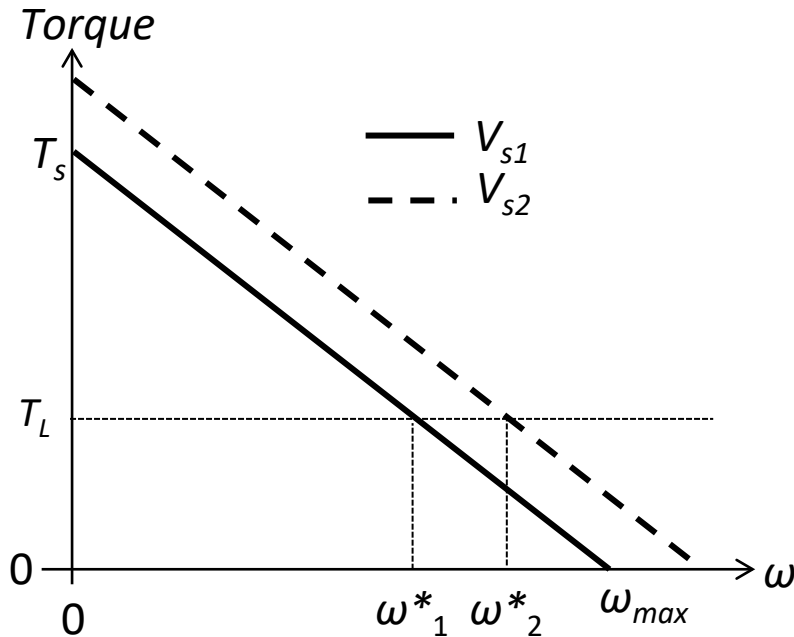
where

J	=	Total Moment of Inertia
J_m	=	Moment of Inertia of the motor
J_L	=	Moment of Inertia of the load
J_L / N^2	=	Reflected load inertia
N	=	transmission reduction of N:1

The angular acceleration: $\alpha_m = N \alpha_L$

where α_m = angular acceleration of the motor
 α_L = angular acceleration of the load

Mechanical Time Constant of PMDC Motor



- If supply voltage suddenly increases from V_{s1} to V_{s2} , the speed won't suddenly change from ω^*_1 to ω^*_2 .
- The transition between two speeds when supply voltage suddenly changes is governed by the mechanical time constant (τ_m); it is the time required for the speed to rise from its old steady-state value to 63% of its new steady-state value.
- $\tau_m = R \cdot J_m / (K_b \cdot K_t)$ where J_m is the inertia of the motor.



Exercise 4

From the provided information, calculate the mechanical time constant and frictional torque in the bearings of the highlighted motor. Compare the calculated time constant with the provided time constant. Also, can you tell if the provided no-load speed is theoretical or actual?

		118693	118694	118695	118696	118697	118698	118699	118700	118701
Motor Data										
Values at nominal voltage										
1 Nominal voltage	V	1.8	2.4	3	3.2	4.5	4.8	7.2	9	12
2 No load speed	rpm	4990	6360	6890	6270	6740	5700	6890	6740	7130
3 No load current	mA	23.5	25.4	23	18.6	14.8	10.8	9.57	7.4	6.05
4 Nominal speed	rpm	4320	5510	5820	4930	5050	3630	4810	4630	5030
5 Nominal torque (max. continuous torque)	mNm	2.39	2.5	2.89	3.41	4.48	5.61	5.54	5.48	5.48
6 Nominal current (max. continuous current)	A	0.72	0.72	0.72	0.72	0.72	0.711	0.566	0.438	0.348
7 Stall torque	mNm	15.5	16.9	17.3	15.2	17.4	15.2	18.1	17.4	18.6
8 Starting current	A	4.53	4.71	4.19	3.13	2.74	1.9	1.82	1.37	1.16
9 Max. efficiency	%	86	86	86	85	86	86	86	86	86
Characteristics										
10 Terminal resistance	Ω	0.397	0.51	0.715	1.02	1.64	2.53	3.95	6.56	10.3
11 Terminal inductance	mH	0.0207	0.0227	0.0302	0.0415	0.0711	0.113	0.174	0.284	0.452
12 Torque constant	mNm/A	3.43	3.58	4.13	4.84	6.34	7.99	9.92	12.7	16
13 Speed constant	rpm/V	2790	2660	2310	1970	1510	1190	962	753	597
14 Speed / torque gradient	rpm/mNm	323	379	400	415	391	378	383	389	386
15 Mechanical time constant	ms	5.84	5.71	5.56	5.46	5.36	5.31	5.29	5.29	5.27
16 Rotor inertia	gcm ²	1.73	1.44	1.33	1.26	1.31	1.34	1.32	1.3	1.3



Exercise 4: Solution

② From the provided information, calculate the mechanical time constant and frictional torque in the bearings of the highlighted motor. Compare the calculated time constant with the provided time constant. Is the provided no-load speed theoretical or actual?

$$\textcircled{1} T_m = \frac{J_m R}{K_t K_b} = \frac{J_m R}{K_t^2} = \frac{1.39 \text{ gcm}^2 \times \frac{1 \text{ kg}}{1000 \text{ g}} \times \frac{1 \text{ m}^2}{(100 \text{ cm})^2} \times 10^{-3} \Omega}{(16 \times 10^{-3})^2 (\text{Nm/A})^2}$$

$$\Rightarrow T_m = 5.23 \times 10^{-3} \text{ s} = 5.23 \text{ ms}$$

② No load current is due to friction
 $i_f = 6.05 \text{ mA}$

$$T_f = K_t i_f = 16 \text{ mNm/A} \times 6.05 \times 10^{-3} \text{ A}$$

$$T_f = 0.0968 \text{ mNm}$$

$$\omega_{\max} = \frac{T_s}{K} = \frac{18.6}{\left(\frac{1}{386}\right)} = 7179.6 \text{ rpm}$$

7179.6 rpm > 7130 rpm (value in Table)

Motor Data			118701
Values at nominal voltage			
1	Nominal voltage	V	12
2	No load speed	rpm	7130
3	No load current	mA	6.05
4	Nominal speed	rpm	5030
5	Nominal torque (max. continuous torque)	mNm	5.48
6	Nominal current (max. continuous current)	A	0.348
7	Stall torque	mNm	18.6
8	Starting current	A	1.16
9	Max. efficiency	%	86
Characteristics			
10	Terminal resistance	Ω	10.3
11	Terminal inductance	mH	0.452
12	Torque constant	mNm/A	16
13	Speed constant	rpm/V	597
14	Speed / torque gradient	rpm/mNm	386
15	Mechanical time constant	ms	5.27
16	Rotor inertia	gcm ²	1.3

Power Model: Steady State

At steady-state:

Recall: $(V_s - V_b) = i_m R$

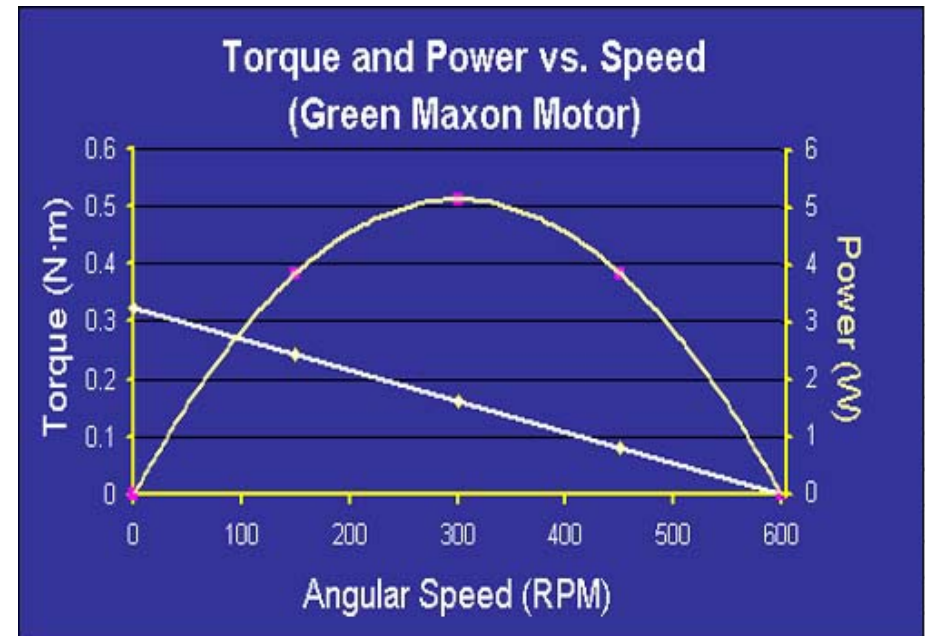
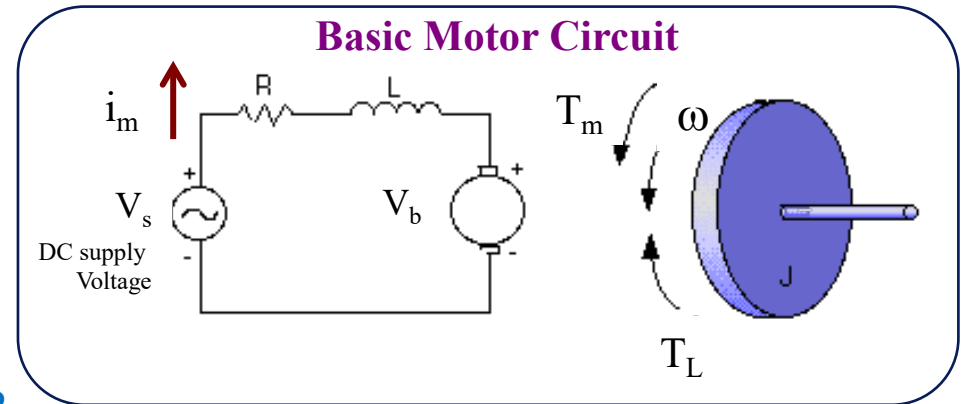
Rewriting: $V_s = K_b \omega + i_m R$

Power is voltage * current

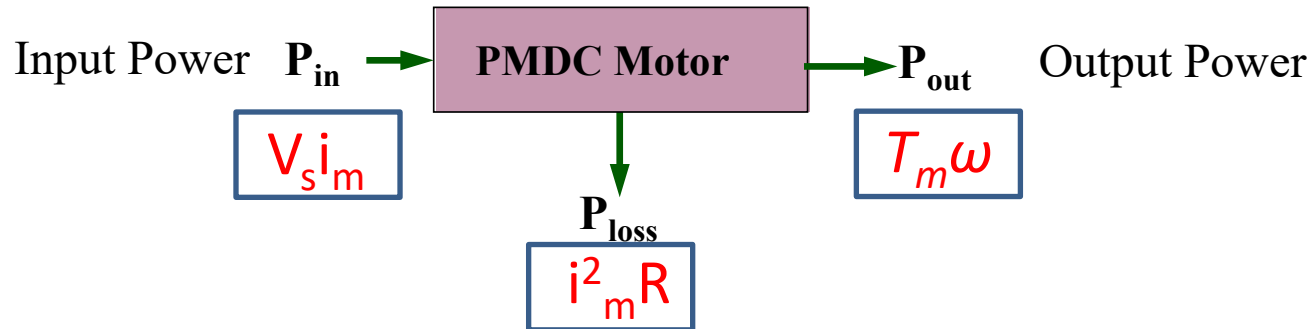
$$i_m V_s = i_m K_b \omega + i_m^2 R = i_m K_T \omega + i_m^2 R$$

this can be simplified to steady-state model:

$$\underbrace{i_m V_s}_{\substack{P_{\text{input}} \\ \text{(Electrical)}}} = \underbrace{T_m \omega}_{\substack{P_{\text{out}} \\ \text{(Mechanical)}}} + \underbrace{R i_m^2}_{\substack{P_{\text{lost}} \\ \text{(Thermal)}}}$$



Power of PMDC Motor



- Derivations of expressions of P_{in} , P_{out} and P_{loss} as functions of ω

- $P_{in} = V_s i_m = V_s \frac{T_m}{K_t} = V_s \frac{(T_s - K\omega)}{K_t} \quad (1) \quad ; \quad V_s \frac{(T_s R)}{K_t} \quad (2) \quad ; \quad K = \frac{K_t^2}{R^2} \quad (3)$

Put (2) and (3) into (1):

$$P_{in} = \frac{T_s(T_s - K\omega)}{K}$$

- $P_{out} = T_m \omega = (T_s - K\omega) \cdot \omega$

$$P_{out} = T_s \omega - K\omega^2$$

- $P_{loss} = i_m^2 R = \frac{T_m^2 \cdot R}{K_t^2} \rightarrow P_{loss} = \frac{(T_s - K\omega)^2}{K}$

Maximum Power of PMDC Motor

$$P_{\text{out}} = T_s \omega - K\omega^2$$

Maximum power of motor occurs at:

$$\frac{dP_{\text{out}}}{d\omega} = 0 = T_s - 2K\omega \rightarrow \omega_{\text{opt}} = \frac{T_s}{2K}$$

$$\left\{ \text{Recall: } \frac{T_s}{K} = \omega_{\text{max}} \right\}$$

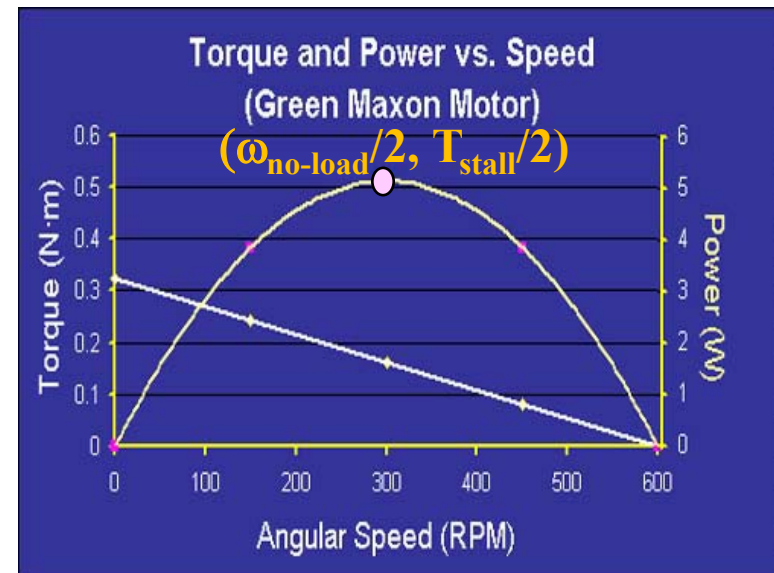
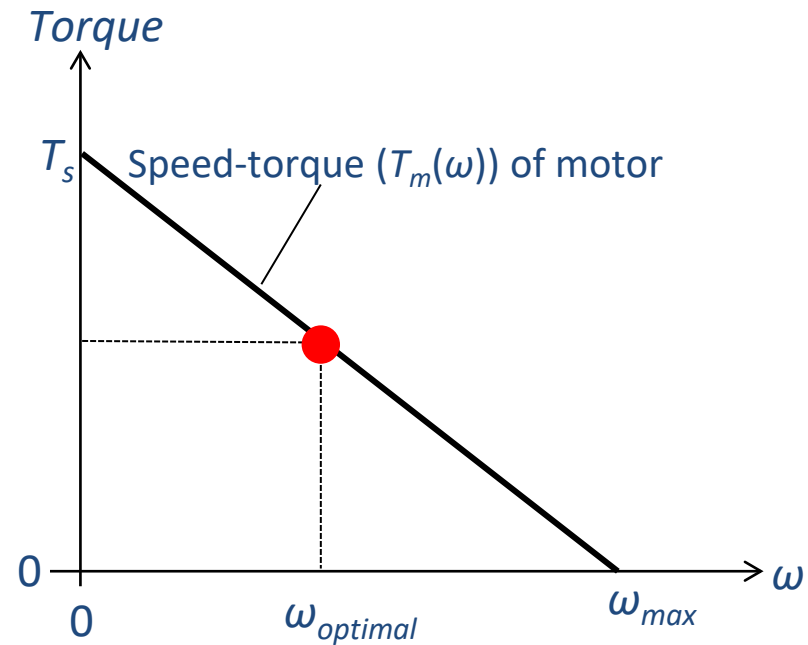
$$\omega_{\text{opt}} = \frac{\omega_{\text{max}}}{2}$$

The resulting Maximum power is:

$$\text{Substituting in } \omega_{\text{opt}} = \frac{T_s}{2K}$$

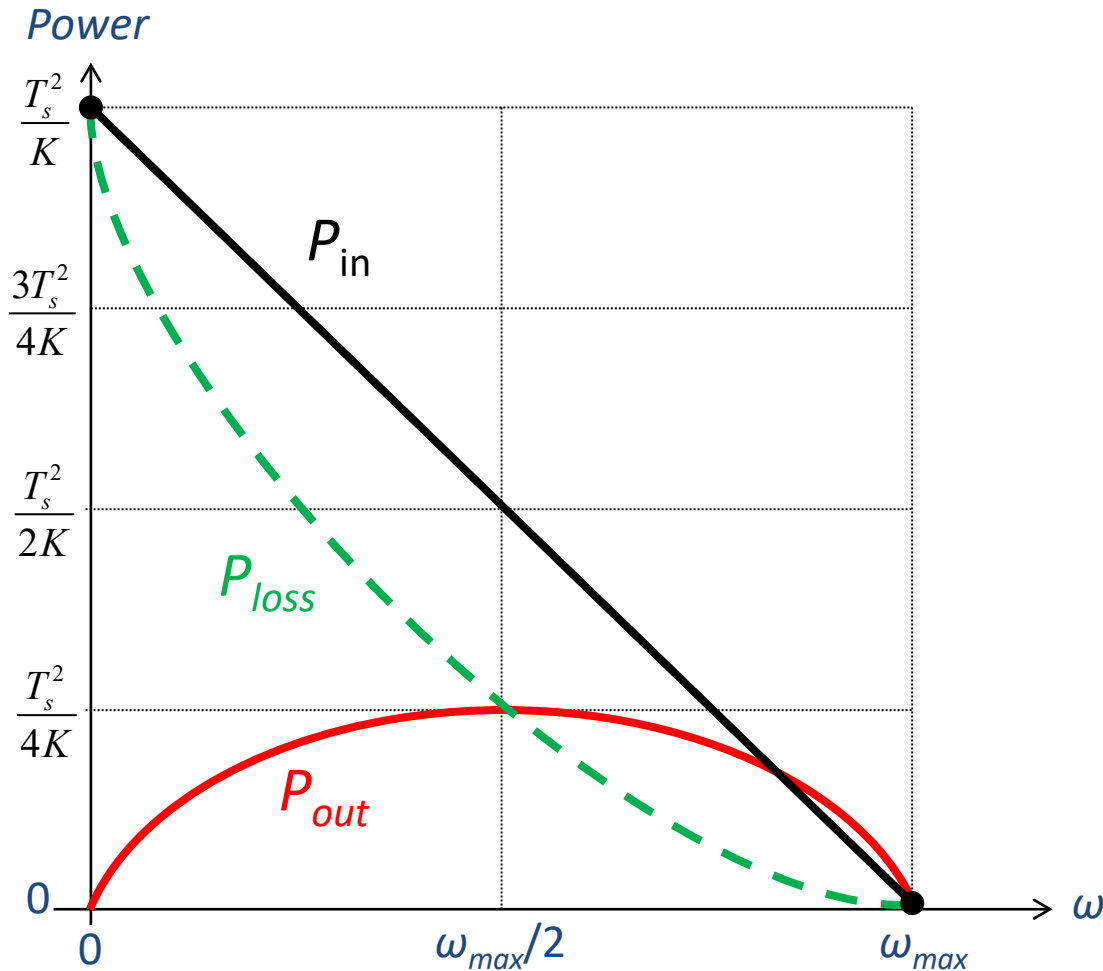
$$\text{to output power } P_{\text{out}} = T_s \omega - K\omega^2$$

$$P_{\text{out max}} = \frac{T_s^2}{2K} - \frac{KT_s^2}{4K^2} = \frac{T_s^2}{4K}$$



Exercise 5:

Sketch the expressions of P_{in} , P_{out} and P_{loss} as functions of ω below



$$P_{in} = \frac{T_s(T_s - K\omega)}{K}$$

$$P_{out} = T_s\omega - K\omega^2$$

$$P_{loss} = \frac{(T_s - K\omega)^2}{K}$$

Theoretical and Actual Efficiencies of PMDC Motor

- The theoretical efficiency (η) only considers**electrical losses**.....; however, the actual efficiency (η_{act}) considers**both electrical & frictional losses**.....

$$(1) \quad \eta = \frac{P_{out}}{P_{in}} = \frac{T_m \omega}{V_S i_m} = \frac{i_m K_t \omega}{V_S i_m} = \frac{\omega}{V_S / K_t} = \boxed{\frac{\omega}{\omega_{max}}}$$

Maximum efficiency at $\eta = 1$ OR $\omega = \omega_{max}$

$$(2) \quad \eta_{act} = \frac{P_{out}}{P_{in}} = \frac{(T_m - T_f) \omega}{V_S i_m} = \frac{(T_s - K \omega - T_f) \omega}{\frac{V_S}{K_t} (T_s - K \omega)}$$

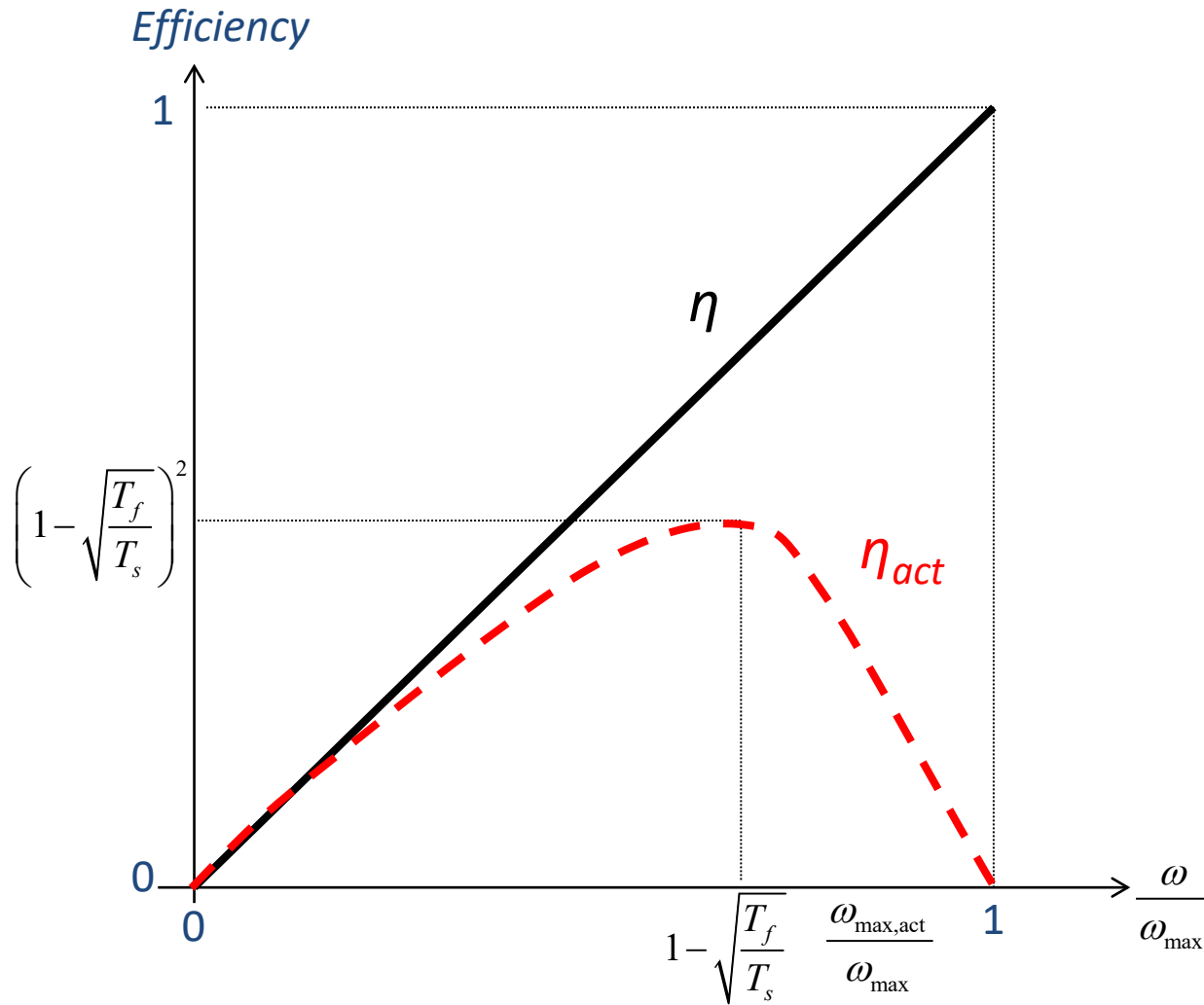
→ Simplify and take $d\eta_{act}/dr = 0$ (where $r = \omega / \omega_{max}$)

Maximum efficiency

$$r_{opt} = \frac{\omega_{opt}}{\omega_{max}} = 1 - \sqrt{\frac{T_f}{T_s}}$$

$$\eta_{act,max} = \left(1 - \sqrt{\frac{T_f}{T_s}} \right)^2$$

Sketch of Theoretical and Actual Efficiencies



Exercise 6

Show that the maximum actual efficiency of the highlighted motor is 86%

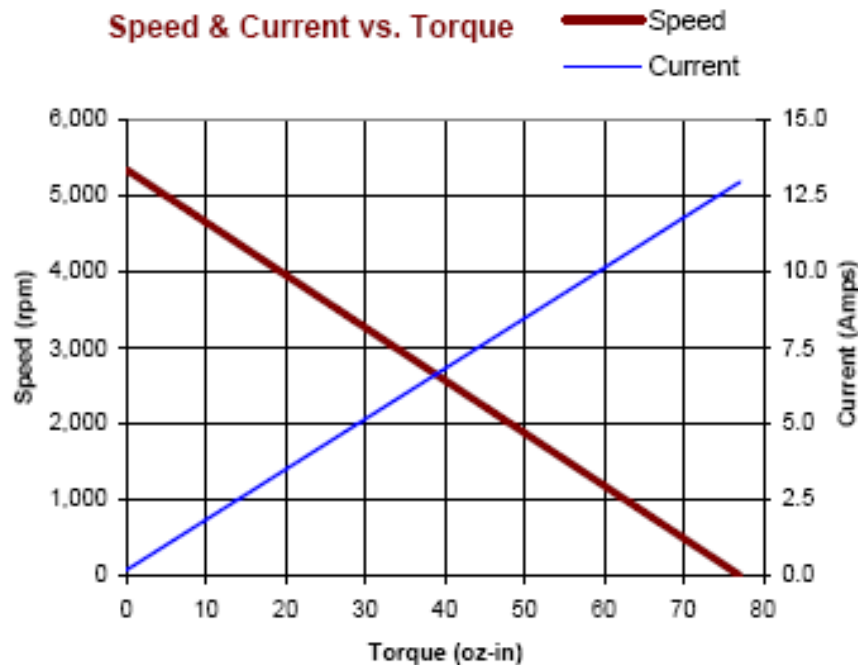
$$\eta_{\text{act,max}} = \left(1 - \sqrt{\frac{T_f}{T_s}}\right)^2 = \left(1 - \sqrt{\frac{0.0968}{18.6}}\right)^2 = 0.86$$

		118693	118694	118695	118696	118697	118698	118699	118700	118701	
Motor Data											
Values at nominal voltage											
1	Nominal voltage	V	1.8	2.4	3	3.2	4.5	4.8	7.2	9	12
2	No load speed	rpm	4990	6360	6890	6270	6740	5700	6890	6740	7130
3	No load current	mA	23.5	25.4	23	18.6	14.8	10.8	9.57	7.4	6.05
4	Nominal speed	rpm	4320	5510	5820	4930	5050	3630	4810	4630	5030
5	Nominal torque (max. continuous torque)	mNm	2.39	2.5	2.89	3.41	4.48	5.61	5.54	5.48	5.48
6	Nominal current (max. continuous current)	A	0.72	0.72	0.72	0.72	0.72	0.711	0.566	0.438	0.348
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9	Max. efficiency	%	86	86	86	85	86	86	86	86	86
Characteristics											
10	Terminal resistance	Ω	0.397	0.51	0.715	1.02	1.64	2.53	3.95	6.56	10.3
11	Terminal inductance	mH	0.0207	0.0227	0.0302	0.0415	0.0711	0.113	0.174	0.284	0.452
12	Torque constant	mNm/A	3.43	3.58	4.13	4.84	6.34	7.99	9.92	12.7	16
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15	Mechanical time constant	ms	5.84	5.71	5.56	5.46	5.36	5.31	5.29	5.29	5.27
16	Rotor inertia	gcm ²	1.73	1.44	1.33	1.26	1.31	1.34	1.32	1.3	1.3

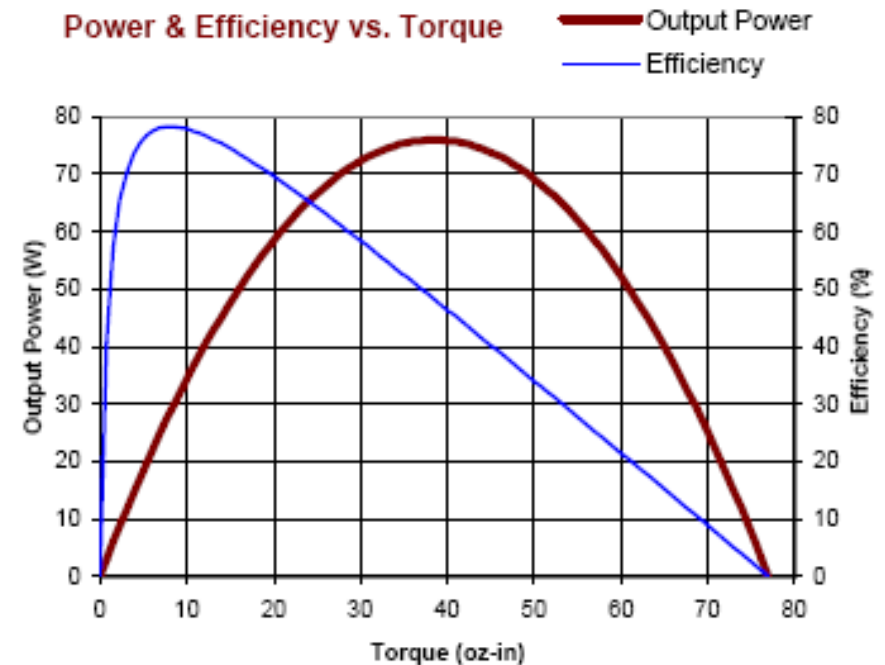
Power and Efficiency Graphs



Speed & Current vs. Torque



Power & Efficiency vs. Torque

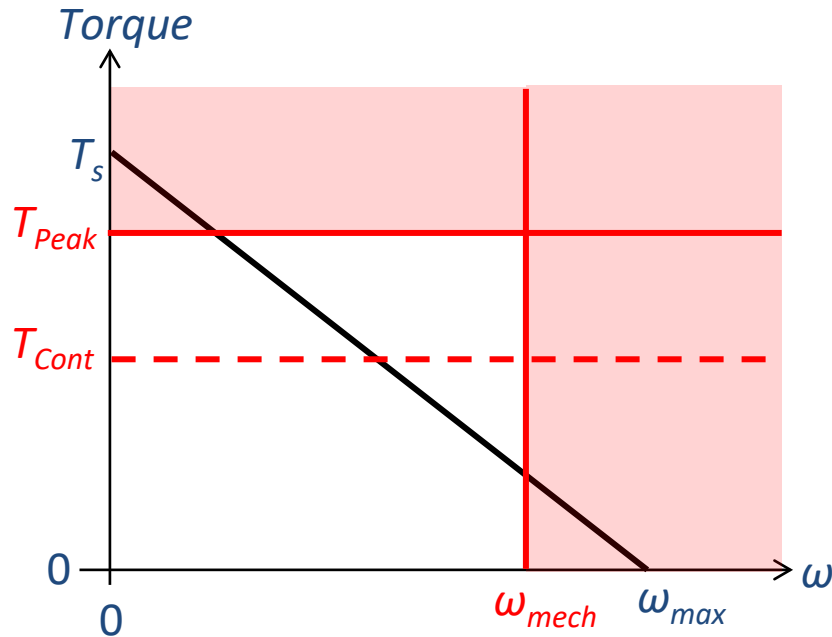


ME 350 Puzzle

Which of the following is the best operating speed to drive a motor?

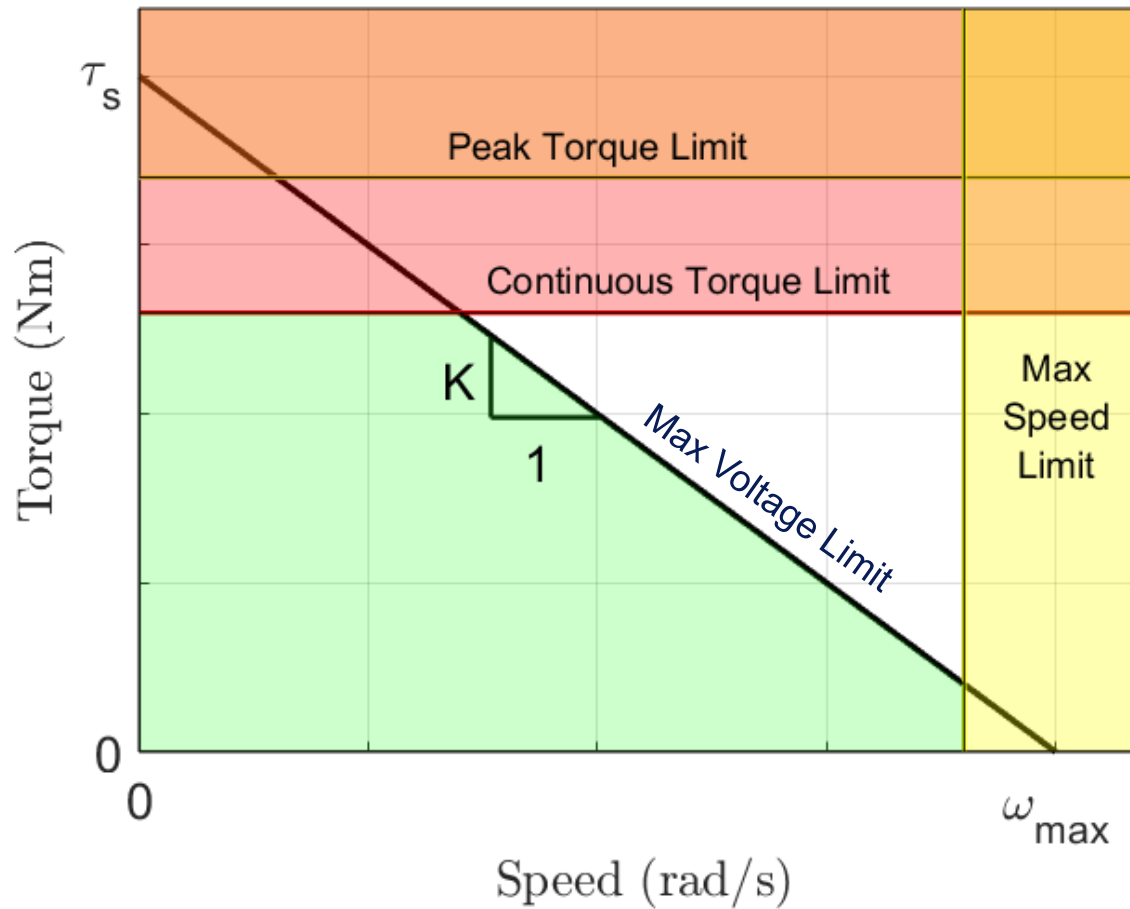
- A) At maximum efficiency
- B) At maximum output power
- C) Not sure

Limits (Constraints) on PMDC



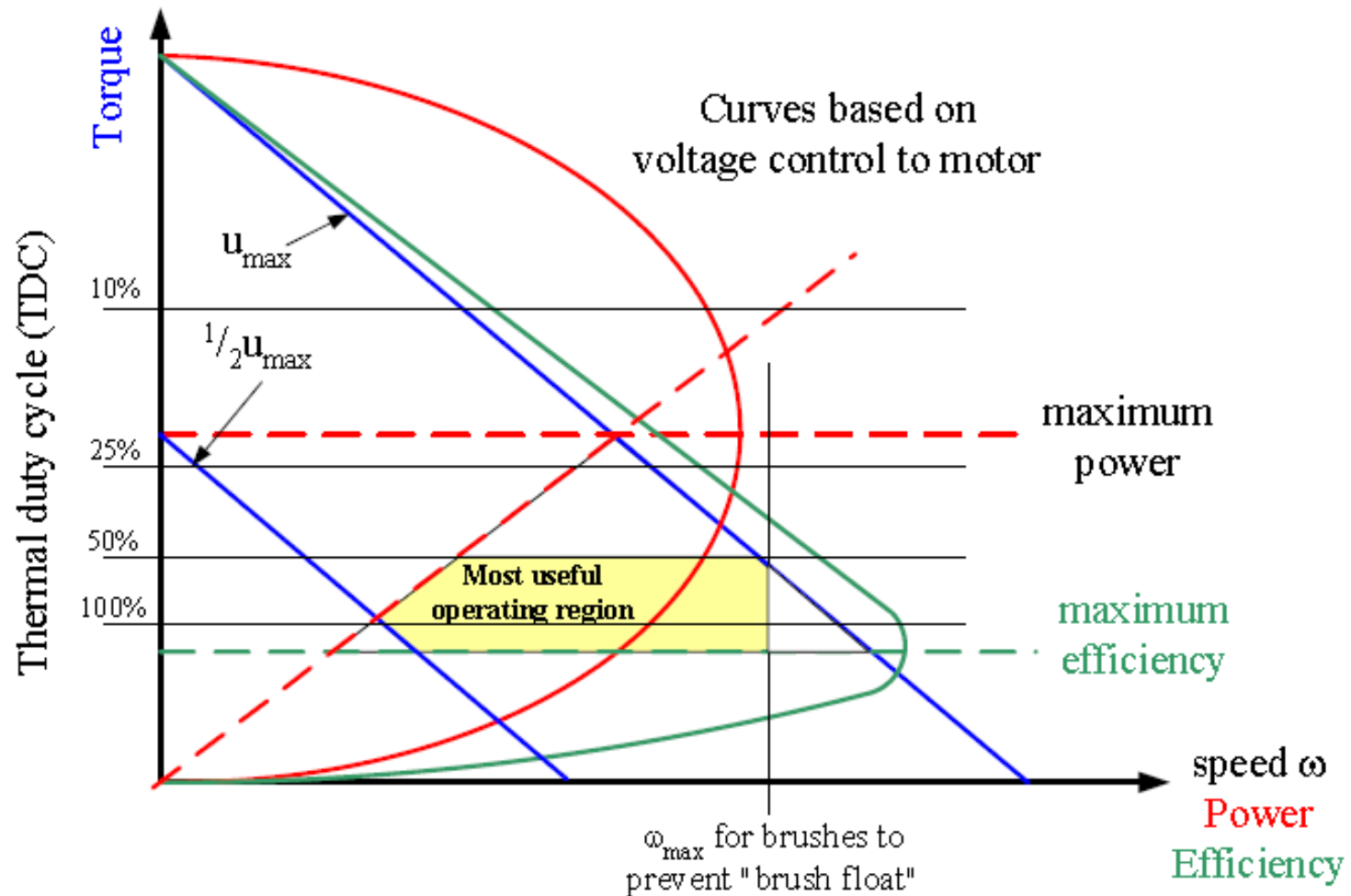
- Theoretically, the speed and torque capabilities of a PMDC can be raised without limits
- However, in practice the following limits must be considered:
 - Motor failure speed (ω_{mech}): The speed beyond which:
the motor fails mechanically
 - Peak torque limit (T_{peak}): The torque beyond which ...
the motor overheats, even for short time periods
 - Continuous torque limit (T_{cont}): The torque beyond which ...
the motor overheats, at long time periods
- The supply voltage must be selected such that these limits are not exceeded.

Motor Operation Limits



Key Motor Curves

Important: There are multiple competing performance objectives



OPTIMIZATION FOR KEY DRIVERS

Optimal Performance

There are traditionally multiple solutions, thus one tries to seek the best one or “optimal”.

To do this, employ Calculus:

- To find a maximum or minimum, take the partial derivative of the function with respect to the variable of interest.
- Set equal to zero
- Solve for the variable of interest to find maximum/minimum

Optimal Operating Torque

One common desire is to set the operating point to transfer as much power as possible.

DC Motor Example:

- Motor curve is $T = T_o + k \omega$ (using the gradient constant)
- Power = speed (in radians per second) x Torque

$$P(\omega) = \omega \times (T_o + k \omega) = T_o \omega + k \omega^2$$

To find the maximum power take the derivative with respect to speed to get

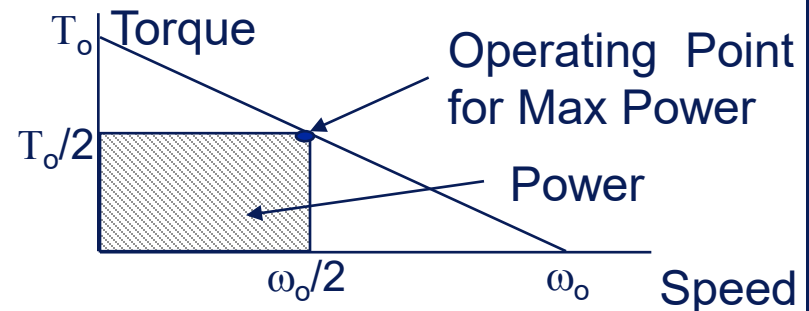
$$\delta P(\omega) / \delta \omega = (T_o + 2 k \omega)$$

Setting this equal to zero and solving for speed gives the maximum power or

$$\omega = -T_o / 2 k = T_o / (2 T_o / \omega_o) = \omega_o / 2$$

for this point the torque is $T = T_o / 2$

Thus, the maximum power is transferred when the torque and the speed are one half of their maximum value.



Start-up Torque (Repeat)

When a motor changes speed, such as at start-up, it undergoes acceleration and the moment of inertia can have a significant impact. The system response time is slowed by the moment of inertia whereas the steady-state speed is unaffected. Sometimes the torque required for start-up is higher than the operating torque and is the main design driver.

Start-up Torque of Motor: $T = J \alpha_m$ or
 $T = (J_m + (J_L / N^2)) \times (N \alpha_L)$

The Moment of Inertia: $J = J_m + (J_L / N^2)$

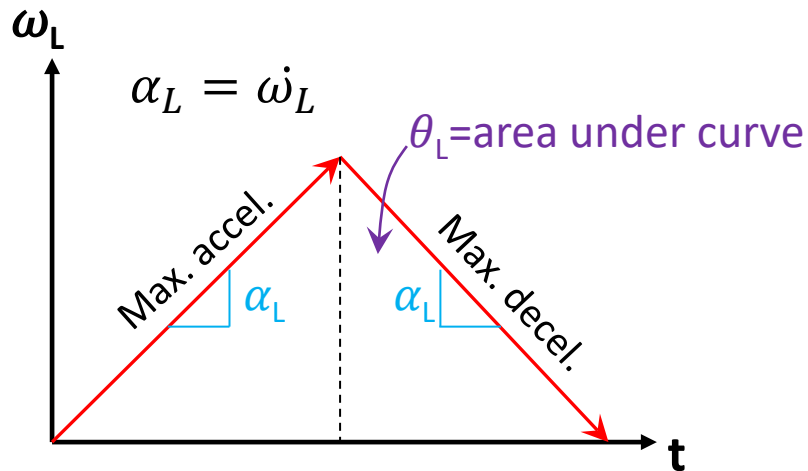
where

J	=	Total Moment of Inertia
J_m	=	Moment of Inertia of the motor
J_L	=	Moment of Inertia of the load
J_L / N^2	=	Reflected load inertia
N	=	transmission reduction of N:1

The angular acceleration: $\alpha_m = N \alpha_L$

where α_m = angular acceleration of the motor
 α_L = angular acceleration of the load

What is the Optimal Transmission Ratio?



Optimal N must maximize ω_L

$$N = \omega_m / \omega_L \rightarrow \omega_m = N \omega_L$$

$$\Rightarrow \dot{\omega}_m = N \dot{\omega}_L \quad \dots(1)$$

$$\text{Recall: } N = (T_L / \eta T_m) \quad \dots(2)$$

To accelerate, we must overcome the inertia of the motor and load:

$$T_m = \underbrace{J_m \dot{\omega}_m}_{\text{Torque to move motor inertia (i.e., } F = ma)} + \underbrace{T_L / N \eta}_{\text{Torque to move load}} \quad \dots(3)$$

$$\& \quad T_L = J_L \dot{\omega}_L \quad \dots(4)$$

$\Rightarrow$ Sub (1), (2), and (4) into (3)

$$T_m = \left(J_m N + \frac{1}{N \eta} J_L \right) \underbrace{\dot{\omega}_L}_{\alpha_L} \quad \dots(5)$$

What is the Optimal Transmission Ratio?

$$T_m = \left(J_m N + \frac{1}{N\eta} J_L \right) \underbrace{\dot{\omega}_L}_{\alpha_L} \quad \dots (5)$$

We want to select N that **maximizes** α_L

Re-write equation (5):

$$\alpha_L = \frac{T_m}{J_m N + \frac{1}{N\eta} J_L}$$

$$\frac{d\alpha_L}{dN} = 0 \rightarrow N = \sqrt{\frac{J_L}{\eta J_m}} \quad \text{For } \eta = 1 \rightarrow \text{(100\% efficiency)}$$

$$N = \sqrt{\frac{J_L}{J_m}}$$

For this transmission ratio the torque is

$$T = 2 \alpha_L N J_m = 2 \alpha_L (J_L J_m)^{0.5}$$

The current is

$$i = T / K_T = 2 \alpha_L (J_L J_m)^{0.5} / K_T$$

The optimal gear ratio has the reflected load of inertia equal the inertia of the motor.
Doing this is called inertial matching.

Minimum Power Dissipation⁴

In addition to best transmission, a designer many times wishes to minimize the power dissipation.

Power dissipation (losses), P is

$$P = i^2 r = (4 \alpha_L^2 J_L J_m r) / K_T^2$$

this can be rewritten as

$$P = (4 \alpha_L^2 J_L) \times [(J_m r) / K_T^2]$$

The second term in the square brackets includes all the motor-dependent parameters and is called the electromechanical time constant, T_m (

$$T_m = [(J_m r) / K_T^2]$$

Thus, power dissipation is minimized by choosing a motor with the shortest electromechanical time constant.

Important – this is NOT the motor torque in previous slides – be careful

Minimum Temperature⁴

A designer also wishes to minimize the temperature rise.

Every motor has a torque level T_c which is known as the continuous or rated torque. This is the largest torque that can be generated continuously by the motor. This implies that the motor may carry a continuous current of

$$i_c = T_c / K_T$$

and the motor can dissipate power at the level of P_c

$$P_c = i_c^2 r = (T_c^2 r) / K_T^2$$

To minimize the temperature rise, select the motor for which the ratio between the power dissipation, P , and the allowed continuous power, P_c , is minimum or the ratio:

$$P / P_c = (4 \alpha_L^2 J_L) \times [J_m / T_c^2]$$

The only motor- dependent term are in the square bracket. The optimal motor is the one in which this term is minimized. The inverse of this term is called

$$\text{The power rate} = T_c^2 / J_m$$

The optimal motor is the one in which the power rate is *maximized*. This motor will have the minimum temperature rise.